

MERIDIAN-1 DATA CENTRE, HALL A

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SECTION 21 22 00 — CLEAN-AGENT FIRE SUPPRESSION

21 22 00 Part 1 — GENERAL

1.1 SUMMARY. Total-flooding clean-agent suppression for the Hall A data hall, UPS rooms, and battery rooms.

1.2 REFERENCES. NFPA 2001, UL 2166, IS 15493.

1.3 SUBMITTALS. Hydraulic calculations, nozzle listing data, agent quantity calculations, room integrity test procedure.

21 22 00 Part 2 — PRODUCTS

2.1 AGENT. FK-5-1-12 (fluoroketone). Recycled or reclaimed agent is not permitted.

2.2 DISCHARGE TIME. Agent discharge to 95 percent of design concentration shall complete within 10 seconds per NFPA 2001. ^[3]

2.3 DESIGN CONCENTRATION. Minimum design concentration 5.6 percent for Class C hazards at the protected enclosure's minimum ambient temperature.

2.4 ENCLOSURE INTEGRITY. Peak positive pressure during discharge shall not exceed 25 Pa with venting as designed. A door-fan integrity test per NFPA 2001 Annex C is required for each protected enclosure prior to acceptance; retention time not less than 10 minutes.

2.5 STORAGE. Cylinder charging pressure 24.8 bar (360 psi) at 21 degrees C, super-pressurised with nitrogen.

Footnote [3]: For protected volumes exceeding 2,000 cubic metres, a discharge time not exceeding 12 seconds is acceptable where extended-discharge nozzles covered by the manufacturer's UL 2166 listing are provided and hydraulic calculations demonstrate 95 percent concentration within 12 seconds.

21 22 00 Part 3 — EXECUTION

3.1 TESTS. Room integrity door-fan test results and puff test records are quality records and prerequisites to Level 4 testing.